Anderson localisation with self-avoiding-walks

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Geboren am 21. November 2004 in Bologna

14th August 2026

Bachelor's Thesis Mathematics

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Zusammenfassung

Wir führen die Benutzung einiger LaTeX-Befehle und -Pakete anhand von Beispielen vor. Die Verwendung der hier empfohlenen Lösungen ist, mit Ausnahme des Titelblatts, nicht verpflichtend, sollte Ihre Arbeit aber erheblich vereinfachen.

Dieses Dokument spiegelt den Stand von 2021 wider; wenn Sie es im Jahr 2030 oder später noch lesen, sollten Sie sich vermutlich nach Alternativen für hier vorgestellte Lösungen umschauen.

—Pavel Zorin-Kranich

Contents

1 Introduction

The Anderson localization model was introduced by the physicist P.W. Anderson in 1958 to explain the quantum mechanical effects of disorder in materials. Anderson discusses the behaviour of electrons in dirty crystals, which is the quantum mechanical analougue of a random walk in a random environment. The model considers a d-dimensional grid, in which each site represents an atom. From an heuristical point of view, we also consider electrons that jump from atom to atom and are subject to an external random potential, which influences their motion in the material. When these particles cannot jump , it indicates the absence of quantum transport.This phenomenon is called Anderson localisation. This model deals with conditions for which the electrons stay trapped in a specific region and don't generate electric conduction, defining an insulator. This is in sharp contrast to the behaviour in ideal crystals, which are always conductors.
Anderson's work, and that of N.F.Mott and J.H. van Vleck, won the 1977 physics Nobel prize for "their fundamental theoretical investigations of the electronic structure of magnetic and disordered systems". Since then, the study of random operators has become an important field of mathematical physics, which has led to a tremendous amount of research activity and many mathematical results.
The two important methods to prove Anderson localization are the *multiscale analysis method* (MSA), developed by Froehlich and Spencer [] in 1983, and the *fractional moment method* (FMM) by Aizenman and Molchanov in 1993. We will deal with the second one, which is more elementary and gives stronger results on *dynamical localization* [Chapter 2.2].

structure of the thesis

Prerequisites

It is useful to have some familiarity with measure theory, basic probabilistic concepts such as indipendence and continuous distributions, and foundations of the theory of linear operators in Hilbert spaces, up to the spectral theorem for self adjoint operators and consequences (as spectral types).

1.1 The Anderson Model

Transport phenomena in a disordered material are often described via discrete random Schroedinger operator. On the lattice $\mathbb{Z}^d$, we consider the infinite random matrix called the *Hamiltonian*:

$$H_\omega = -\Delta + \lambda V_\omega.$$

It is **Guarda paper diesrtori per info iniziali, mi sembravano scritte bene.**This operator acts on the Hilbert space

$$l^2(\mathbb{Z}^d) = \{u : \mathbb{Z}^d \to \mathbb{C} : \sum_{n \in \mathbb{Z}^d} |u(n)|^2 < \infty\}$$

with inner product $\langle u, v\rangle = \sum_n \overline{u(n)} v(n)$.

1.1.1 The operator

The Descrete Laplacian. The kinetic term $-\Delta$ is the descrete analogue of the usual negative Lalacian $(-\sum_j \partial^2/\partial x_j^2)$. For the functions $u \in l^2(\mathbb{Z}^d)$ it is defined as:

$$-\Delta u(x) := -\sum_{k \in \mathbb{Z}^d, |x-k|=1} u(x) - u(k), \quad k \in \mathbb{Z}^d$$

where $|x - k|$ is the graph difference in $\mathbb{Z}^d$. The descrete Laplacian is per definiton bounded and consequently self-adjoint [for proofs see Appendix A]. Just as the continuum Laplacian the Fourier trnasform blah blah blah.
The spectrum is absolutely continuous and as similarity with the continuus laplacian, it has plane waves as generalized eigenfunctions.

The Potential Term.
... **The disorder strength.**

1.1.2 Self-adjointness

1.1.3 The spectrum of H_ω.

- in $\mathbb{R}$
- bounded $[0, 4d] + supp\mu$
- spectral division (any use of the spectral theorem should be sent to the Appendix)
- RAGE theorem (or maybe directly in the Appendix or chapter 2)

1.2 Main results

In this section one could write first of all the main theorem and the spectral results. starting maybe with:
The aim of this thesis is to prove localisation on the whole spectrum of H_ω
$\dots$
Poi scrivo il teorema finale con spiegazione che la dimostrazione avverrà nel capitolo 5.
potrebbe essere un'idea anche scrivere una piccola parentesi sulla differenza dei rislutati con Frohlich e spencer e con Aizenman e Molchanov.
One of the main results concerns the values of λ

- $\lambda >> 1$ and $\lambda = 0$
- general case for $d = 1, d \geq 2$

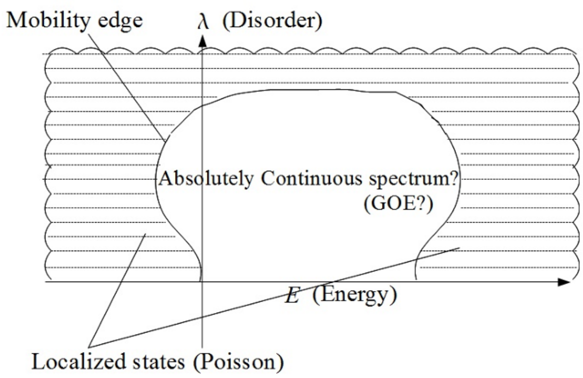

Figure 1: Aizenman figure

2 Localisation properties

We understood what localisation means from an heuristical point of view. Mathematically, it can be described from spectral and dynamical properties.

2.1 spectral localisation

...

Definition 2.1. ...

2.2 dynamical localisation

Definition 2.2.

$$\sup_{t\in\mathbb{R}} |||X|^p e^{-itH_\omega}\phi|| < \infty \qquad \text{almost surely} \ldots$$

2.2.1 exponential localisation

Definition 2.3.

$$\mathbb{E}\left(\sup_{t\in\mathbb{R}} |\langle e_j, e^{-itH_\omega} e_k\rangle|\right) \leq Ce^{-\mu|j-k|}$$

for all φ with compact support.

...

2.3 From dynamical to spectral localisation

Theorem 2.4. *If H shows dynamical localisation in an interval I, then it has pure point spectrum in I*

Proof. ... □

3 Fractional moments

3.1 The green function

Lemma 3.1. *(a priori bound)*

3.2 First localisation criteria

Theorem 3.2. *first important theorem*

3.2.1 Proof without SAW

4 SAW representation

4.1 SAW Definitions

4.2 SAW representation of G

- Theorem of the representation

  1. first prove that $G(x,y) = G(x,x)\sum G(x_1,y)$
  2. then the represantion with the R_N

- put figure of the grid ant the different paths

4.3 Proof with SAW

5 From fractional moments to dynamical localisation

Proposition 5.1.

$$|\Im z|\mathbb{E}\left(|G(x,y;z)|^2\right) \leq C_1\mathbb{E}\left(|G(x,y;z)|^s\right)$$

Theorem 5.2. *Theorem final*